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IT2021 AIML Project – 2026 Jan

Topic Approval Form

1. Project ID: Y2.S2.WD.DS-02-G04

2. Project Title: Web-Based Tour Guide

3. Campus: Malabe

4. WD/WE: WD

5. Group Information:
   Stream: Data Science

	Registration Number	**Student Name**	**Phone Number**	**Signature**
01	IT24400066	Athauda P.A.K. D	0779247253	
02	IT24104301	Karannagoda K.V. R	0764231685	
03	IT24101887	Faizam M.F.M	0761707745	
04	IT24101446	Wijekoon W.M.N.S. B	0768609939	
05	IT24102131	De Silva P.K. N	0760188303	
06	IT24101896	Abeywickrama J. B	0719842955	

6. Client Information / Project Justification
   A. Project with a Client:
      Client Name/ Organization:

      Contact Person & Designation:

      Email:

   B. Projects without a Client:

   Since this project does not have an external client, the topic "AI-Based Web Tour Guide System" was selected based on real-world relevance and learning value.
   Tourism is an important industry, and many travelers struggle to choose suitable tour packages and guides due to lack of personalized recommendations. We can use web development and AI/ML concepts in a real-world setting with this project.
   The project is appropriate for our academic goals because it offers chances to practice full-stack development, teamwork, CRUD operations, and AI/ML integration.

7. A brief description of the problem.

Many tourists face difficulties when planning trips, such as selecting suitable tour packages, finding reliable guides, managing bookings, and getting personalized recommendations. Most traditional tour websites only provide static information and do not adapt to individual user preferences. The proposed system aims to solve this problem by providing a Web-Based Tour Guide System that allows users to browse tourist spots, select tour packages with guides, make bookings, give feedback, and receive smart recommendations. This system's AI/ML component focuses on intelligent location prediction and recommendation. The machine learning model forecasts and suggests appropriate tourist destinations for each user based on past user data, browsing habits, ratings, and preferences saved in tabular (CSV) format. By offering tailored location recommendations, this turns the platform from a static information system into an intelligent application that enhances user experience and system efficacy while assisting travelers in making better travel choices.

8. Main Features of the Proposed System.

List software features. Software features should be functional and meaningful parts of the system, not just simple UI pages. Where possible, they should connect to or use the AI/ML feature.

	Registration Number (Same order as above)	**Name of Feature/s**	**Brief description of feature(s) in point form**
01	IT24101887	User Authentication	• Register/Login as Tourist, Guide, Admin • Role-based access • Profile management
02	IT24101896	Tourist Spot Browsing	• View tourist locations • Read descriptions & reviews • Search and filter spots
03	IT24102131	Tour Package + Guide Selection	• View tour packages • See assigned guides • Select package and guide together • View price and duration
04	IT24400066	Booking Management	• Book selected tours • View booking history • Cancel or reschedule bookings
05	IT24104301	Feedback & Review System	• Submit ratings after tours • Write reviews • View public feedback
06	IT24101446	Payment Management	• Handle tour payments • Store payment status • Generate basic receipts
07		AI Chat Bot	• Q and A

9. AI/ML Feature of the Proposed System.

Feature Name: Smart Tour Package Recommendation

	Registration Number (Same order as above)	Expected individual contribution to the AI/ML feature.
01	IT24400066	Build & run the Genetic Algorithm code (core optimization engine).
02	IT24104301	Add review summaries to chatbot answers.
03	IT24101887	User preferences from profile > feed to chatbot for better answers.
04	IT24101446	Build "Optimize Trip" page UI + show GA results.
05	IT24102131	Chatbot logic for suggesting packages/guides.
06	IT24101896	Prepare spot data for chatbot (embeddings + retrieval).

10. Approval by the Evaluator

Name: Dr. Dharshana Kasthurirathne

Date: 03-02-2026